

Clinical Lab Results Analyzer

Submit lab values, get each one classified as **Critical / Warning / Normal** with an explanation of why it was flagged and a suggested next step — then take the panel away as a PDF, or ask an assistant about it in your own words. **Severity is decided by deterministic code, never by the language model.** The model translates a decision that has already been made.

Pipeline

Step	Runs on	What it decides
1. Classify	MCP tool	Compare each value to its reference range. Decides the severity.
2. Route	MCP tool	Group and order by severity, critical first. Decides the order.
3. Explain	Gemini	One batched call for the whole panel. Decides only the wording.

Features

Feature	What it gives you	Built with
Deterministic classification	Normal / Warning / Critical decided by a fixed threshold comparison — never by the LLM	MCP tool <code>classify_lab_result</code>
Severity routing	Results ordered worst-first, with counts per severity	MCP tool <code>route_by_severity</code>
Reference table	16 tests with aliases, units, and critical thresholds — served to the UI, never hardcoded in it	MCP tool <code>list_reference_ranges</code>
Written explanations	What the test measures, what your result means, causes, urgency, next steps, questions for a doctor	Google Gemini 3.5-flash-lite, one batched call
Audit trail	Every card expands to the literal comparison that fired, the range used, and where it came from	Python; surfaced in the API response
PDF lab report	The panel as a real document: measured tables, row-aware page breaks, Page X of Y, branding	ReportLab, server-side
CSV export	Full result set including ranges and the rule that fired	Browser-side; no round trip
Assistant — about the app	Answers from the repository's own docs and the reference table, with sources shown	Ollama qwen2.5:3b + nomic-embed-text; NumPy cosine similarity
Assistant — about your report	Ask about your own results in plain words; answers quote your numbers back	Ollama qwen2.5:3b — no retrieval, the report is the whole context
Safety guards	Medical questions refused in code; greetings answered without a model	Deterministic Python, before the LLM runs
CSV upload + preview	See exactly what was parsed before anything is classified	FastAPI + python-multipart
Bundled datasets	Four files runnable in one click, including the real Kaggle set	FastAPI
Test-name picker	Names come from the server; a name off the list cannot be submitted	React combobox fed over MCP
Degraded mode	LLM unreachable? Every severity, range and rule still returned	Agent failure policy
Feedback	Stored server-side rather than posted to a mail service with keys in the bundle	FastAPI + JSONL

Tools and libraries

Tool	Role in this project
FastAPI (Python 3.12)	HTTP API, validation, generated OpenAPI docs
MCP over stdio (SDK v2.1)	All clinical logic behind 4 tools, reusable by any MCP client
Google Gemini 3.5-flash-lite	Explanations only. ~4.5× faster than larger flash models, no quality loss here
Ollama qwen2.5:3b	Assistant chat, local. Instruct not reasoning — qwen3:4b took ~60s on CPU
Ollama nomic-embed-text	Assistant embeddings, local, 768-dimension
NumPy	Vector index over 116 chunks. A vector DB would add a service and be slower
ReportLab	PDF generation. A print stylesheet cannot do measured tables or page breaks
React 19 + Vite 8	Four pages, two themes, shared analysis state in the shell
pytest	317 tests, no network and no API key required

Guarantees

Property	Behaviour
MCP unavailable	The request fails (503). Without classification there is nothing trustworthy to return.
LLM unavailable	The request succeeds without explanations. Severity is computed locally and must not be lost to an outage.
Agent isolation	The agent never imports the tool server — every call goes over MCP. A test asserts the boundary.
Assistant locality	Chat and embeddings are local only. No cloud fallback, so lab values never leave the machine.

Validated against the Kaggle dataset *Laboratory Test Results — Anonymized* (CC0-1.0): 27 rows, 0 unclassified, 0 disagreements with the laboratory's own verdict, which is never shown to the classifier. Turkish test names, per-row reference intervals and qualitative urinalysis values are all handled. **Not a medical device** — it compares numbers to published ranges and knows nothing about the person they came from.